

Trijet cross section at large N_c (Massive case)

1 Setup

The cross section for trijet production is given by

$$\frac{d^3 N}{d^3 k_1 d^3 k_2 d^3 k_3} (2\pi) \delta(q^+ - k_1^+ - k_2^+ - k_3^+) = \frac{1}{2} \int_{w, w'} e^{-iq \cdot (w - w')} \langle \gamma_T^i(Q, q^+, w') | \mathcal{N}_q(k_1) \mathcal{N}_{\bar{q}}(k_2) \mathcal{N}_g(k_3) | \gamma_T^i(Q, q^+, w) \rangle_{q\bar{q}g} \quad (1.1)$$

1.1 Kinematics

The virtual photon with the four momentum as $q^\mu = (q^+, -\frac{Q^2}{2q^+}, q)$. Let us define the kinematics in the following: The onshell

Component	Longitudinal momentum fraction	Transverse momentum	mass
Quark	$\zeta = \frac{k_1^+}{q^+}$	k_1	m_f
Anti Quark	$\eta = 1 - \zeta - \xi$	k_2	m_f
Gluon	$\xi = \frac{k_3^+}{q^+}$	k_3	0

denominators are $E_q(k) = \frac{k^2 + m_f^2}{2k^+}$, $E_{\bar{q}}(p) = \frac{p^2 + m_f^2}{2p^+}$, $E_g(p) = \frac{p^2}{2p^+}$.

1.2 Matrix Elements

The matrix elements that are used in this process are as follows:

1. Virtual photon splitting into a quark and anti quark pair

$$\langle q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p) | H_{\gamma \rightarrow q\bar{q}} | \gamma_\lambda(q) \rangle = (2\pi)^3 \delta^{(3)}(q - k - p) \frac{e e_f \delta_{\alpha\beta}}{2\sqrt{2}q^+} \chi_{\lambda_1}^\dagger \left[\frac{2q^\lambda}{q^+} - \sigma^\lambda \frac{\sigma \cdot p + m_f \sigma^3}{p^+} - \frac{\sigma \cdot k - m_f \sigma^3}{k^+} \sigma^\lambda \right] \chi_{\lambda_2} \quad (1.2)$$

rewriting it using the delta function,

$$k^+ = \zeta q^+, p^+ = (1 - \zeta) q^+, \tilde{k} = k - \zeta q$$

we get

$$\left[\frac{2q^\lambda}{q^+} - \frac{\sigma \cdot k - m_f \sigma^3}{k^+} \sigma^\lambda - \sigma^\lambda \frac{\sigma \cdot (q - k) + m_f \sigma^3}{(q^+ - k^+)} \right] = \frac{[(2\zeta - 1) \delta^{\lambda j} + i \epsilon^{\lambda j} \sigma^3] \tilde{k}^j + m_f \sigma^3 \sigma^\lambda}{\zeta (1 - \zeta) q^+}$$

where we used $\sigma^i \sigma^j = \delta^{ij} + i \epsilon^{ij} \sigma^3$ and $\sigma^\lambda \sigma^3 = -\sigma^3 \sigma^\lambda$ and

$$\chi_{\lambda_1}^\dagger [(2\zeta - 1) \delta^{\lambda j} + i \epsilon^{\lambda j} \sigma^3] \chi_{\lambda_2} = \delta_{\lambda_1 \lambda_2} [(2\zeta - 1) \delta^{\lambda j} + 2i \epsilon^{\lambda j} \lambda_1]$$

$$\chi_{\lambda_1}^\dagger \sigma^3 \sigma^\lambda \chi_{\lambda_2} = \delta_{\lambda_2, -\lambda_1} [2\lambda_1 \delta^{\lambda, 1} - i \delta^{\lambda, 2}] \equiv \mu_{\lambda_1 \lambda_2}^\lambda$$

The mass term flips the helicity ($\lambda_2 = -\lambda_1$). The matrix element defined above can be rewritten as follows:

$$\langle q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p) | H_{\gamma \rightarrow q\bar{q}} | \gamma_\lambda(q) \rangle = (2\pi)^3 \delta^{(3)}(q - k - p) \frac{e e_f \delta_{\alpha\beta} \left\{ [(2\zeta - 1) \delta^{\lambda j} + 2i \epsilon^{\lambda j} \lambda_1] \delta_{\lambda_1 \lambda_2} \tilde{k}^j + m_f \mu_{\lambda_1 \lambda_2}^\lambda \right\}}{2\sqrt{2}q^+ \zeta (1 - \zeta)} \quad (1.3)$$

2. Gluon emission from anti - quark

$$\langle q_{\lambda_1}^{\alpha'}(k') \bar{q}_{\lambda_2}^{\beta'}(p') g_{i'}^a(q') | H_{\bar{q}g} | q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p) \rangle = -(2\pi)^6 \delta^{(3)}(k - k') \delta^{(3)}(p - p' - q') \delta_{\alpha' \alpha} \delta_{\lambda_1 \lambda_1'} \frac{g t_{\beta \beta'}^{a'}}{2\sqrt{2}q'^+} \chi_{\lambda_2}^\dagger \left[\frac{2q'^{i'}}{q'^+} - \sigma^{i'} \frac{\sigma \cdot p' + m_f \sigma^3}{p'^+} - \frac{\sigma \cdot p + m_f \sigma^3}{p^+} \sigma^{i'} \right] \chi_{\lambda_2'} \quad (1.4)$$

with $p^+ = (1 - \zeta) q^+$, $p'^+ = \eta q^+$, $q'^+ = \xi q^+$ and $\tilde{p} = q' - \frac{\xi}{(1 - \zeta)} p$. The terms proportional to p cancel and, from here onwards with $\omega_f = \frac{\xi m_f}{(1 - \zeta)}$,

$$\chi_{\lambda}^\dagger [\dots] \chi_{\lambda_2} = \frac{\tau_{\lambda \lambda_2}^{i' n}(\xi, \eta) \tilde{p}^n + \xi \omega_f \mu_{\lambda \lambda_2}^{i'}}{\xi \eta q^+}, \quad m_f \sigma^3 \sigma^{i'} \left(\frac{1}{p'^+} - \frac{1}{p^+} \right) = \frac{\xi^2 m_f}{(1 - \zeta)} \frac{\sigma^3 \sigma^{i'}}{\xi \eta q^+}.$$

3. Instantaneous contribution

$$\langle q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(s) g_j^b(p) | H_{\gamma \rightarrow q\bar{q}g} | \gamma \rangle = -(2\pi)^3 \delta^{(3)}(q - k - p - s) \frac{g e e_f \delta_{\lambda_1 \lambda_2} t_{\alpha\beta}^b}{4\sqrt{q^+ p^+}} \frac{\delta^{ij} + 2i \epsilon^{ij} \lambda_1}{p^+ + s^+} \quad (1.5)$$

1.3 Notations

$$\begin{aligned}\tilde{k} &= k - \zeta q, & \tilde{p} &= p - \frac{\xi}{1-\zeta} (q - k), & \tilde{Q}^2 &= \zeta(1-\zeta) Q^2 \\ \epsilon_f^2 &= \tilde{Q}^2 + m_f^2 = \zeta(1-\zeta) Q^2 + m_f^2, & \omega_f &= \frac{\xi m_f}{(1-\zeta)}, & E_f^2 &= Q^2 + m_f^2 \left(\frac{1}{\zeta} + \frac{1}{\eta} \right), & a &= \frac{\xi \eta}{\zeta(1-\zeta)^2} \\ \varphi_{\lambda_1 \lambda}^{ij}(\zeta) &= \delta_{\lambda_1 \lambda} [(2\zeta - 1)\delta^{ij} + 2i\epsilon^{ij}\lambda_1], & \tau_{\lambda\lambda_2}^{mn}(\xi, \eta) &= \delta_{\lambda\lambda_2} [(2(1-\zeta) - \xi)\delta^{mn} + 2i\xi\epsilon^{mn}\lambda] \\ \mu_{\lambda\lambda'}^i &= \delta_{\lambda', -\lambda} [2\lambda\delta^{i1} - i\delta^{i2}] = \chi_\lambda^\dagger \sigma^3 \sigma^i \chi_{\lambda'}, & \sum_i \mu_{\lambda\lambda'}^i \mu_{\lambda\lambda'}^{i*} &= 2\delta_{\lambda', -\lambda}\end{aligned}$$

Here ϵ_f is the usual mass shifted photon scale. The photon vertex and the anti quark - gluon vertex are:

$$\text{photon : } \varphi_{\lambda_1 \lambda}^{ij}(\zeta) \tilde{k}^j + m_f \mu_{\lambda_1 \lambda}^i, \quad \text{gluon off the antiquark : } \tau_{\lambda\lambda_2}^{mn}(\xi, \eta) \tilde{p}^n + \xi \omega_f \mu_{\lambda\lambda_2}^m$$

The energy denominator is as follows:

$$\begin{aligned}E_q + E_{\bar{q}} - E_\gamma &= \frac{\tilde{k}^2 + \tilde{Q}^2 + m_f^2}{2\zeta(1-\zeta)q^+} = \frac{\tilde{k}^2 + \epsilon_f^2}{2\zeta(1-\zeta)q^+} \\ E_{q_1} + E_{\bar{q}_1} + E_{g_1} - E_\gamma &= \frac{(1-\zeta)(\tilde{p}^2 + \omega_f^2)}{2\xi\eta q^+} + \frac{(\tilde{k}^2 + \epsilon_f^2)}{2\zeta(1-\zeta)q^+} = \frac{\zeta(1-\zeta)^2(\tilde{p}^2 + \omega_f^2) + \xi\eta(\tilde{k}^2 + \epsilon_f^2)}{2\zeta(1-\zeta)\xi\eta q^+} \\ E_{q_1} + E_{\bar{q}_1} + E_{g_1} - E_q - E_{\bar{q}} &= \frac{p^2}{2q^+\xi} + \frac{(q-k-p)^2 + m_f^2}{2q^+\eta} - \frac{(q-k)^2 + m_f^2}{2q^+(1-\zeta)} = \frac{[(1-\zeta)p - \xi(q-k)]^2}{2\xi(1-\zeta)\eta q^+} + \frac{m_f^2}{2q^+} \left(\frac{1}{\eta} - \frac{1}{1-\zeta} \right) \\ &= \frac{(1-\zeta)^2 \tilde{p}^2 + \xi^2 m_f^2}{2\xi(1-\zeta)\eta q^+}\end{aligned}\tag{1.6}$$

Simplifying it we get

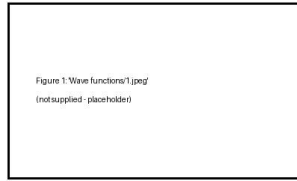
$$E_{q_1} + E_{\bar{q}_1} + E_{g_1} - E_q - E_{\bar{q}} = \frac{(1-\zeta)(\tilde{p}^2 + \omega_f^2)}{2\xi\eta q^+}$$

Also, for instantaneous term involving gluon emission is given by

$$\zeta(1-\zeta)^2 \omega_f^2 + \xi\eta\epsilon_f^2 = \zeta(1-\zeta)\xi\eta E_f^2, \quad E_f^2 = Q^2 + \frac{m_f^2}{\zeta} + \frac{m_f^2}{\eta}\tag{1.7}$$

2 Outgoing component of wave function through LCPT

1. Anti quark emits a gluon before the shock wave



$$|\gamma\rangle_{q\bar{q}g}^{(1)} = |q_2\bar{q}_2g_2\rangle \langle q_2\bar{q}_2g_2| S |q_1\bar{q}_1g_1\rangle \times \frac{\langle q_1\bar{q}_1g_1| H_{\bar{q}g} |q\bar{q}\rangle}{E_{\bar{q}_1} + E_{q_1} + E_{g_1} - E_\gamma} \times \frac{\langle q\bar{q}| H_{\gamma \rightarrow q\bar{q}} |\gamma\rangle}{E_q + E_{\bar{q}} - E_\gamma}\tag{2.1}$$

$$\begin{aligned}|\gamma\rangle_{q\bar{q}g}^{(1)} &= \int d^3k'' d^3p'' d^3r d^3k' d^3p' d^3q' d^3k d^3p |q_{\lambda_1'}^{\alpha''}(k'') \bar{q}_{\lambda_2'}^{\beta''}(p'') g_j^b(r)\rangle \langle q_{\lambda_1'}^{\alpha''}(k'') \bar{q}_{\lambda_2'}^{\beta''}(p'') g_j^b(r)| S |q_{\lambda_1}^{\alpha'}(k') \bar{q}_{\lambda_2}^{\beta'}(p') g_{i'}^{a'}(q')\rangle \times \\ &\frac{\langle q_{\lambda_1'}^{\alpha'}(k') \bar{q}_{\lambda_2'}^{\beta'}(p') g_{i'}^{a'}(q')| H_{\bar{q}g} |q_{\lambda_1}^{\alpha}(k) \bar{q}_{\lambda_2}^{\beta}(p)\rangle}{E_{\bar{q}}(p') + E_q(k') + E_g(q') - E_\gamma} \times \frac{\langle q_{\lambda_1}^{\alpha}(k) \bar{q}_{\lambda_2}^{\beta}(p)| H_{\gamma \rightarrow q\bar{q}} |\gamma_\lambda(q)\rangle}{E_{\bar{q}}(p) + E_q(k) - E_\gamma(q, Q)}\end{aligned}\tag{2.2}$$

We will first evaluate the matrix elements without scattering (i.e., $S = 1$). With the massive vertices and energy denominators of Sec. 1, we get the following

$$\begin{aligned}|\gamma_i(Q, q^+, q)\rangle_{q\bar{q}g}^{(1a)} &= -\frac{1}{(2\pi)^6} \int d^2\tilde{k} d^2\tilde{p} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^{\alpha}(\zeta q^+, k) \bar{q}_{\lambda_2}^{\beta}(\eta q^+, q - k - p) g_m^a(\xi q^+, p)\rangle \frac{e e_f g q^+ \zeta(1-\zeta)}{2\sqrt{\xi}(\tilde{k}^2 + \epsilon_f^2)} \\ &\frac{[\varphi_{\lambda_1 \lambda}^{ij}(\zeta) \tilde{k}^j + m_f \mu_{\lambda_1 \lambda}^i] [\tau_{\lambda\lambda_2}^{mn}(\xi, \eta) \tilde{p}^n + \xi \omega_f \mu_{\lambda\lambda_2}^m] t_{\alpha\beta}^a}{[\zeta(1-\zeta)^2(\tilde{p}^2 + \omega_f^2) + \xi\eta(\tilde{k}^2 + \epsilon_f^2)]}\end{aligned}\tag{2.3}$$

However, Wilson lines can be obtained only if the wave function is expressed in transverse coordinates. Therefore, we perform a Fourier transform to rewrite the wave function in transverse coordinate space. Performing Fourier transform for transverse co-ordinates,

$$|q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(q-k-p) g_m^a(p)\rangle = \int_{x,y,z} e^{ik \cdot x} e^{ip \cdot z} e^{i(q-k-p) \cdot y} |q_{\lambda_1}^\alpha(x) \bar{q}_{\lambda_2}^\beta(y) g_m^a(z)\rangle \quad (2.4)$$

Rewriting k, p in terms of $\tilde{k}$ and $\tilde{p}$ respectively, we get

$$\begin{aligned} e^{i[\zeta x + \eta y + \xi z] \cdot q} e^{-i(y-z) \cdot \tilde{p}} e^{i[(1-\zeta)x - \eta y - \xi z] \cdot \frac{\tilde{k}}{(1-\zeta)}} &= e^{i(\zeta x + \eta y + \xi z) \cdot q} e^{-i(y-z) \cdot \tilde{p}} e^{i\left(x - \frac{\eta y + \xi z}{(1-\zeta)}\right) \cdot \tilde{k}} \\ &= e^{i w \cdot q} e^{-i Y \cdot \tilde{p}} e^{i R \cdot \tilde{k}}. \end{aligned} \quad (2.5)$$

The transverse coordinates of the photon (w) and the intermediate antiquark (y' , before emitting the gluon) are fixed by centre of energy conservation at the two vertices:

$$w = \zeta x + (1-\zeta) y' = \zeta x + \eta y + \xi z, \quad (2.6)$$

and

$$y' = \frac{\eta y + \xi z}{(1-\zeta)}. \quad (2.7)$$

The relative distances are defined as

$$R = x - y'; \quad Y = y - z$$

where we used

$$(1-\zeta)x - \eta y - \xi z = (1-\zeta)(x - y') = (1-\zeta)R$$

We will use the following Fourier transforms:

$$\int \frac{d^2 k}{2\pi} \frac{d^2 p}{2\pi} \frac{k^i p^j}{k^2 [ak^2 + bp^2]} e^{ik \cdot X + ip \cdot Z} = -\frac{X^i Z^j}{Z^2 [bX^2 + aZ^2]} \quad (2.8)$$

With masses the two denominators contain two different scales ϵ_f and ω_f and the transform is no longer a single Bessel function. We need four transforms, one with each helicity path. Define

$$(1-\zeta) \int \frac{d^2 \tilde{k}}{2\pi} \frac{d^2 \tilde{p}}{2\pi} \frac{e^{i\tilde{k} \cdot R} e^{-i\tilde{p} \cdot Y} \{\tilde{k}^j \tilde{p}^n, \tilde{k}^j, \tilde{p}^n, 1\}}{(\tilde{k}^2 + \epsilon_f^2) [\zeta(1-\zeta)^2 (\tilde{p}^2 + \omega_f^2) + \xi \eta (\tilde{k}^2 + \epsilon_f^2)]} = \left\{ R^j Y^n \mathcal{G}^{(11)}, iR^j \mathcal{G}^{(10)}, -iY^n \mathcal{G}^{(01)}, \mathcal{G}^{(00)} \right\} \quad (2.9)$$

Using Schwinger parametrization, we get

$$\frac{1}{(\tilde{k}^2 + \epsilon_f^2) [\zeta(1-\zeta)^2 (\tilde{p}^2 + \omega_f^2) + \xi \eta (\tilde{k}^2 + \epsilon_f^2)]} = \frac{1}{\xi \eta} \int_0^\infty du \int_0^\infty dv e^{-(u+v)(\tilde{k}^2 + \epsilon_f^2) - \frac{v}{a}(\tilde{p}^2 + \omega_f^2)} \quad (2.10)$$

where $a = \frac{\xi \eta}{[\zeta(1-\zeta)^2]}$, K_ν is the Modified Bessel function of the second kind. Using the Gaussian integrals and including massive terms, we define the following notations, with x as a Feynman parameter.

$$\begin{aligned} \mathcal{G}^{(\alpha\beta)}(R, Y) &= \frac{a^\beta}{2\zeta(1-\zeta)} \int_0^1 \frac{dx}{x^{1+\beta}} \left(\frac{A_x}{\Delta_x} \right)^{\frac{(\alpha+\beta)}{2}} K_{\alpha+\beta}(\sqrt{A_x \Delta_x}), \alpha, \beta \in \{0, 1\} \\ A_x &= \epsilon_f^2 + \frac{x \omega_f^2}{a} = \zeta(1-\zeta) Q^2 + m_f^2 \left(1 + \frac{x \zeta \xi}{\eta} \right) \\ \Delta_x &= R^2 + \frac{a Y^2}{x} \end{aligned}$$

Explicitly

$$\mathcal{G}^{(11)} = \frac{a}{2\zeta(1-\zeta)} \int_0^1 \frac{dx}{x^2} \frac{A_x}{\Delta_x} K_2(\sqrt{A_x \Delta_x}) \quad (2.11)$$

$$\mathcal{G}^{(10)} = \frac{1}{2\zeta(1-\zeta)} \int_0^1 \frac{dx}{x} \sqrt{\frac{A_x}{\Delta_x}} K_1(\sqrt{A_x \Delta_x}) \quad (2.12)$$

$$\mathcal{G}^{(01)} = \frac{a}{2\zeta(1-\zeta)} \int_0^1 \frac{dx}{x^2} \sqrt{\frac{A_x}{\Delta_x}} K_1(\sqrt{A_x \Delta_x}) \quad (2.13)$$

$$\mathcal{G}^{(00)} = \frac{1}{2\zeta(1-\zeta)} \int_0^1 \frac{dx}{x} K_0(\sqrt{A_x \Delta_x}) \quad (2.14)$$

Now, implementing all these steps in the outgoing component of wave function, we get

$$|\gamma_i(Q, q^+, q)\rangle_{q\bar{q}g}^{(1a)} = -\frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, x) \bar{q}_{\lambda_2}^\beta(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\zeta t_{\alpha\beta}^a}{\sqrt{\xi}} \delta^{(2)}(w-c) \times \Psi_{\lambda_1\lambda_2}^{im}[\mathcal{G}] \quad (2.15)$$

where

$$\Psi_{\lambda_1\lambda_2}^{im}[\mathcal{G}] = \sum_{\lambda} \left[\varphi_{\lambda_1\lambda}^{ij}(\zeta) \tau_{\lambda\lambda_2}^{mn}(\xi, \eta) R^j Y^n \mathcal{G}^{(11)} + i \xi \omega_f \varphi_{\lambda_1\lambda}^{ij}(\zeta) \mu_{\lambda\lambda_2}^m R^j \mathcal{G}^{(10)} \right. \\ \left. - i m_f \mu_{\lambda_1\lambda}^i \tau_{\lambda\lambda_2}^{mn}(\xi, \eta) Y^n \mathcal{G}^{(01)} + m_f \xi \omega_f \mu_{\lambda_1\lambda}^i \mu_{\lambda\lambda_2}^m \mathcal{G}^{(00)} \right] \quad (2.16)$$

$$\mathcal{D}^2(x, y, z, \zeta, \xi) = \zeta(1-\zeta)R^2 + \frac{\xi\eta Y^2}{(1-\zeta)} = \zeta\eta(x-y)^2 + \zeta\xi(x-z)^2 + \eta\xi(y-z)^2$$

$$\mu_{\lambda\lambda'}^i = \delta_{\lambda', -\lambda} [2\lambda\delta^{i1} - i\delta^{i2}] = \chi_\lambda^\dagger \sigma^3 \sigma^i \chi_\lambda$$

$$\epsilon_f^2 = \tilde{Q}^2 + m_f^2 = \zeta(1-\zeta)Q^2 + m_f^2, \quad \omega_f = \frac{\xi m_f}{(1-\zeta)}, \quad E_f^2 = Q^2 + m_f^2 \left(\frac{1}{\zeta} + \frac{1}{\eta} \right), \quad a = \frac{\xi\eta}{\zeta(1-\zeta)^2}$$

$$\varphi_{\lambda_1\lambda}^{ij}(\zeta) = \delta_{\lambda_1\lambda} [(2\zeta-1)\delta^{ij} + 2i\epsilon^{ij}\lambda_1], \quad \tau_{\lambda\lambda_2}^{mn}(\xi, \eta) = \delta_{\lambda\lambda_2} [(2(1-\zeta)-\xi)\delta^{mn} + 2i\xi\epsilon^{mn}\lambda]$$

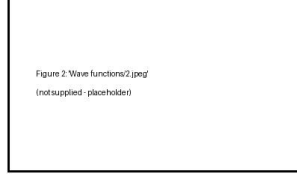
and

$$c = \zeta x + \eta y + \xi z$$

is the centre of energy of trijet system. Attaching Wilson lines, we get

$$|\gamma_i(Q, q^+, q)\rangle_{q\bar{q}g}^{(1a)} = -\frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, x) \bar{q}_{\lambda_2}^\beta(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\zeta}{\sqrt{\xi}} \delta^{(2)}(w-c) \times \Psi_{\lambda_1\lambda_2}^{im}[\mathcal{G}](R, Y) \\ [U^{ab}(z) V(x) t^b V^\dagger(y) - t^a]_{\alpha\beta} \quad (2.17)$$

2. Anti quark emits a gluon after the shock wave



$$|\gamma\rangle_{q\bar{q}g}^{(2)} = |q_2\bar{q}_2g_2\rangle \langle q_2\bar{q}_2g_2| S |q_1\bar{q}_1g_1\rangle \times \frac{\langle q_1\bar{q}_1g_1| H_{\bar{q}g} |q\bar{q}\rangle}{E_{\bar{q}_1} + E_{q_1} - E_{q_2} - E_{\bar{q}_2} - E_{g_2}} \times \frac{\langle q\bar{q}| H_{\gamma \rightarrow q\bar{q}} |\gamma\rangle}{E_q + E_{\bar{q}} - E_\gamma} \quad (2.18)$$

$$|\gamma\rangle_{q\bar{q}g}^{(2)} = \int d^3k'' d^3p'' d^3r d^3k' d^3p' d^3k d^3p |q_{\lambda_1'}^{\alpha''}(k'') \bar{q}_{\lambda_2'}^{\beta''}(p'') g_j^b(r)\rangle \frac{\langle q_{\lambda_1'}^{\alpha''}(k'') \bar{q}_{\lambda_2'}^{\beta''}(p'') g_j^b(r) | H_{\bar{q}g} |q_{\lambda_1}^{\alpha'}(k') \bar{q}_{\lambda_2}^{\beta'}(p')\rangle}{E_q(k') + E_{\bar{q}}(p') - E_q(k'') - E_{\bar{q}}(p'') - E_g(r)} \times \\ \langle q_{\lambda_1'}^{\alpha'}(k') \bar{q}_{\lambda_2'}^{\beta'}(p') | S |q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p)\rangle \times \frac{\langle q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p) | H_{\gamma \rightarrow q\bar{q}} |\gamma_\lambda(q)\rangle}{E_{\bar{q}}(p) + E_q(k) - E_\gamma(q, Q)} \quad (2.19)$$

The only difference between the first and the second wave function component is the energy denominator and the Wilson line structure. The changes are as follows:

$$\frac{1}{E_{q\bar{q}g} - E_\gamma} \rightarrow \frac{1}{E_{q\bar{q}} - E_{q\bar{q}g}}$$

Equivalently,

$$\frac{1}{\zeta(1-\zeta)^2(\tilde{p}^2 + \omega_f^2) + \xi\eta(\tilde{k}^2 + \epsilon_f^2)} \rightarrow -\frac{1}{\zeta(1-\zeta)^2(\tilde{p}^2 + \omega_f^2)}$$

Also, the Fourier transform which we will be using here is the following:

$$\int \frac{d^2\tilde{k}}{2\pi} e^{i\tilde{k}\cdot R} \frac{\{\tilde{k}^j, 1\}}{\tilde{k}^2 + \epsilon_f^2} = \left\{ \frac{iR^j}{R} \epsilon_f K_1(\epsilon_f R), K_0(\epsilon_f R) \right\}$$

and

$$\int \frac{d^2\tilde{p}}{2\pi} e^{-i\tilde{p}\cdot Y} \frac{\{\tilde{p}^n, 1\}}{(\tilde{p}^2 + \omega_f^2)} = \left\{ -\frac{iY^n}{Y} \omega_f K_1(\omega_f Y), K_0(\omega_f Y) \right\}$$

With the same normalisation, we get

$$(1 - \zeta) \int \frac{d^2 \tilde{k}}{2\pi} \frac{d^2 \tilde{p}}{2\pi} e^{i\tilde{k} \cdot R} e^{-i\tilde{p} \cdot Y} \frac{\left\{ \tilde{k}^j \tilde{p}^n, \tilde{k}^j, \tilde{p}^n, 1 \right\}}{(\tilde{k}^2 + \epsilon_f^2) \zeta (1 - \zeta)^2 (\tilde{p}^2 + \omega_f^2)} = \left\{ R^j Y^n \mathcal{A}^{(11)}, i R^j \mathcal{A}^{(10)}, -i Y^n \mathcal{A}^{(01)}, \mathcal{A}^{(00)} \right\} \quad (2.20)$$

where:

$$\begin{aligned} \mathcal{A}^{(11)} &= \frac{\epsilon_f K_1(\epsilon_f R)}{\zeta(1 - \zeta) R} \frac{\omega_f K_1(\omega_f Y)}{Y} \\ \mathcal{A}^{(10)} &= \frac{\epsilon_f K_1(\epsilon_f R)}{\zeta(1 - \zeta) R} K_0(\omega_f Y) \\ \mathcal{A}^{(01)} &= \frac{K_0(\epsilon_f R)}{\zeta(1 - \zeta)} \frac{\omega_f K_1(\omega_f Y)}{Y} \\ \mathcal{A}^{(00)} &= \frac{K_0(\epsilon_f R)}{\zeta(1 - \zeta)} K_0(\omega_f Y) \end{aligned}$$

The Wilson line will not have any component of gluon because the gluons are not getting scattered as the gluon is emitted after the scattering.

$$[U^{ab}(z) V(x) t^b V^\dagger(y) - t^a]_{\alpha\beta} \rightarrow [V(x) V^\dagger(y') t^a - t^a]_{\alpha\beta}$$

Substituting all these changes in eq. (2.17), we get

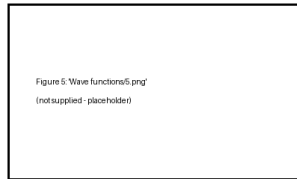
$$\begin{aligned} |\gamma(w)\rangle_{q\bar{q}g}^{(2)} &= + \frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, x) \bar{q}_{\lambda_2}^\beta(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\zeta}{\sqrt{\xi}} \delta^{(2)}(w - c) \times \\ &\quad \Psi_{\lambda_1 \lambda_2}^{im}[\mathcal{A}](R, Y) \times [V(x) V^\dagger(y') t^a - t^a]_{\alpha\beta} \end{aligned} \quad (2.21)$$

where

$$\begin{aligned} \Psi_{\lambda_1 \lambda_2}^{im}[\mathcal{A}] &= \sum_\lambda \left[\varphi_{\lambda_1 \lambda}^{ij}(\zeta) \tau_{\lambda \lambda_2}^{mn}(\xi, \eta) R^j Y^n \mathcal{A}^{(11)} + i \xi \omega_f \varphi_{\lambda_1 \lambda}^{ij}(\zeta) \mu_{\lambda \lambda_2}^m R^j \mathcal{A}^{(10)} \right. \\ &\quad \left. - i m_f \mu_{\lambda_1 \lambda}^i \tau_{\lambda \lambda_2}^{mn}(\xi, \eta) Y^n \mathcal{A}^{(01)} + m_f \xi \omega_f \mu_{\lambda_1 \lambda}^i \mu_{\lambda \lambda_2}^m \mathcal{A}^{(00)} \right] \end{aligned} \quad (2.22)$$

In the massless case Eq. (2.21) follows from Eq. (2.17) by the formal substitution $(b) = (a)|_{y,z \rightarrow y'}$ of the paper (keeping the prefactor $1/Y^2$). With masses this is no longer literally true: the functions $\mathcal{A}^{(\alpha\beta)}$ are different functions from $\mathcal{G}^{(\alpha\beta)}$ (after the shock wave the gluon is emitted from an on-shell antiquark and sees only the dead-cone scale ω_f). We keep the label $[y, z \rightarrow y']$ for this term, meaning: Wilson lines at y' and $\mathcal{G} \rightarrow \mathcal{A}$. The short-distance behaviour is the same, $Y^2 \mathcal{G}^{(11)}, Y^2 \mathcal{A}^{(11)} \xrightarrow{Y \rightarrow 0} \frac{\epsilon_f K_1(\epsilon_f R)}{\zeta(1 - \zeta) R}, Y^2 \mathcal{G}^{(01)}, Y^2 \mathcal{A}^{(01)} \xrightarrow{Y \rightarrow 0} \frac{K_0(\epsilon_f R)}{\zeta(1 - \zeta)}$, and $\mathcal{G}^{(10)} - \mathcal{A}^{(10)}, \mathcal{G}^{(00)} - \mathcal{A}^{(00)}$ stay finite, so the UV cancellation between Eq. (2.17) and Eq. (2.21) works as in the massless case.

3. Instantaneous contribution (Anti quark emission of gluon when an instantaneous vertex acts before the shockwave)



$$|\gamma\rangle_{q\bar{q}g}^{(1 \text{ inst.})} = |q\bar{q}g\rangle \langle q\bar{q}g| S |q\bar{q}g\rangle \frac{\langle q\bar{q}g| H_{\gamma \rightarrow q\bar{q}g} |\gamma\rangle}{E_{q\bar{q}g} - E_\gamma} \quad (2.23)$$

Using the relevant matrix we get (In momentum space - without Wilson lines)

$$\begin{aligned} |\gamma\rangle_{q\bar{q}g}^{(1 \text{ inst.})} &= + \frac{1}{(2\pi)^6} \int d^2 \tilde{k} d^2 \tilde{p} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, k) \bar{q}_{\lambda_2}^\beta(\eta q^+, q - k - p) g_m^a(\xi q^+, p)\rangle \frac{e e_f g q^+ \zeta}{2\sqrt{\xi}} \\ &\quad \frac{(\delta^{im} + 2i \epsilon^{im} \lambda_1) \delta_{\lambda_1 \lambda_2} t_{\alpha\beta}^a}{\tilde{k}^2 + \epsilon_f^2 + \frac{\zeta(1-\zeta)^2}{\xi \eta} (\tilde{p}^2 + \omega_f^2)} \end{aligned} \quad (2.24)$$

Performing Fourier transform and inserting $S - 1$ component, we get the following

$$(1 - \zeta) \int \frac{d^2 \tilde{k}}{2\pi} \frac{d^2 \tilde{p}}{2\pi} e^{i\tilde{k} \cdot R} e^{-i\tilde{p} \cdot Y} \frac{1}{\zeta(1 - \zeta)^2 (\tilde{p}^2 + \omega_f^2) + \xi \eta (\tilde{k}^2 + \epsilon_f^2)} = \frac{E_f K_1(E_f \mathcal{D})}{\mathcal{D}} = \mathcal{G}_E \quad (2.25)$$

The outgoing component of wave function is obtained as follows:

$$|\gamma\rangle_{q\bar{q}g}^{(1\text{ inst.})} = +\frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |\bar{q}_{\lambda_2}^\beta(\eta q^+, y), g_m^a(\xi q^+, z), q_{\lambda_1}^\alpha(\zeta q^+, x)\rangle \frac{\zeta \eta \sqrt{\xi}}{(1-\zeta)} \delta^{(2)}(w-c) (\delta^{im} + 2i \epsilon^{im} \lambda_1) \delta_{\lambda_1 \lambda_2} \frac{E_f K_1(E_f \mathcal{D})}{\mathcal{D}} [U^{ab}(z)V(x)t^b V^\dagger(y) - t^a]_{\alpha\beta} \quad (2.26)$$

This has the same Wilson line structure as the regular i.e., Eq. (2.17). There is no instantaneous term after the shock wave (the gluon emission after the shock wave is a single vertex), so the instantaneous term has no $[y, z \rightarrow y']$ counterpart (in the massless note this was expressed by the factor $\frac{Y^2}{R \cdot Y}$ of the effective vertex, which vanishes for $y, z \rightarrow y'$). Now, we have all the wave functions corresponding to anti quark emission. All four anti quark emission can be written as a single expression with an effective vertex. The effective vertex is written as

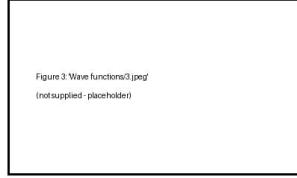
$$\mathcal{M}_{1\lambda_1\lambda_2}^{im}(x, y, z, \zeta, \xi) = \Psi_{\lambda_1\lambda_2}^{im}[\mathcal{G}](R, Y) - \delta_{\lambda_1\lambda_2} \frac{\xi \eta}{(1-\zeta)} (\delta^{im} + 2i \epsilon^{im} \lambda_1) \mathcal{G}_E \quad (2.27)$$

$$\mathcal{M}_{2\lambda_1\lambda_2}^{im}(x, y, z, \zeta, \xi) = \Psi_{\lambda_1\lambda_2}^{im}[\mathcal{A}](R, Y) \quad (2.28)$$

All four anti quark emission are then written using a single expression:

$$|\gamma^i(w)\rangle_{q\bar{q}g} = -\frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |\bar{q}_{\lambda_2}^\beta(\eta q^+, y), g_m^a(\xi q^+, z), q_{\lambda_1}^\alpha(\zeta q^+, x)\rangle \frac{\zeta}{\sqrt{\xi}} \delta^{(2)}(w-c) \left[\mathcal{M}_{1\lambda_1\lambda_2}^{im}(x, y, z, \zeta, \xi) [U^{ab}(z)V(x)t^b V^\dagger(y) - t^a]_{\alpha\beta} - \mathcal{M}_{2\lambda_1\lambda_2}^{im} [V(x) V^\dagger(y')t^a - t^a]_{\alpha\beta} \right] \quad (2.29)$$

4. Quark emits a gluon before the shock wave



When we move from anti quark emission to quark emission, we will have to perform the following changes from anti quark wave functions:

- $\zeta \leftrightarrow \eta$
- $x \leftrightarrow y$
- $\alpha \leftrightarrow \beta$
- $\lambda_1 \leftrightarrow \lambda_2$
- Change overall sign
- $y' \rightarrow x' = \frac{\zeta x + \xi z}{\zeta + \xi}$
- $R \rightarrow y - x' = -\tilde{R}$
- $Y \rightarrow x - z = X$
- $\varphi_{\lambda_1\lambda}^{ij}(\zeta) \rightarrow \varphi_{\lambda_2\lambda}^{ij*}(\eta) = -\varphi_{\lambda\lambda_2}^{ij}(\zeta + \xi)$
- $\tau_{\lambda\lambda_2}^{mn}(\xi, \eta) \rightarrow \tau_{\lambda\lambda_1}^{mn*}(\xi, \zeta) = \tau_{\lambda_1\lambda}^{nm}(\xi, \zeta)$
- $D(x, y, z, \zeta, \xi) \rightarrow D(y, x, z, \eta, \xi)$
- after the shock wave (massless limit): $\mathcal{D} \rightarrow \sqrt{\eta(1-\eta)} \tilde{R}$, i.e. $\tilde{Q} \rightarrow \sqrt{\eta(1-\eta)} Q$
- $m_f \mu_{\lambda_1\lambda}^i \rightarrow m_f \mu_{\lambda\lambda_2}^i$
- $\xi \omega_f \mu_{\lambda\lambda_2}^m \rightarrow \xi \tilde{\omega}_f \mu_{\lambda_1\lambda}^m$ with $\tilde{\omega}_f = \frac{\xi m_f}{(1-\eta)}$
- $\epsilon_f^2 \rightarrow \tilde{\epsilon}_f^2 = \eta(1-\eta) Q^2 + m_f^2$, $a \rightarrow \tilde{a} = \frac{\xi \zeta}{\eta(1-\eta)^2}$
- $A_x \rightarrow \tilde{A}_x = \eta(1-\eta) Q^2 + m_f^2 \left(1 + \frac{x\eta\xi}{\zeta}\right)$ and $\Delta_x \rightarrow \tilde{\Delta}_x = \tilde{R}^2 + \frac{\tilde{a} X^2}{x}$
- $\mathcal{G}^{(\alpha\beta)}(R, Y) \rightarrow \tilde{\mathcal{G}}^{(\alpha\beta)}(\tilde{R}, X)$ and $\mathcal{A}^{(\alpha,\beta)}(R, Y) \rightarrow \tilde{\mathcal{A}}^{(\alpha\beta)}(\tilde{R}, X)$ also, $\zeta(1-\zeta) \rightarrow \eta(1-\eta)$

$$|\gamma\rangle_{q\bar{q}g}^{(3)} = |q_2 \bar{q}_2 g_2\rangle \langle q_2 \bar{q}_2 g_2| S |q_1 \bar{q}_1 g_1\rangle \times \frac{\langle q_1 \bar{q}_1 g_1 | H_{qg} | q\bar{q} \rangle}{E_{\bar{q}_1} + E_{q_1} + E_{g_1} - E_\gamma} \times \frac{\langle q\bar{q} | H_{\gamma \rightarrow q\bar{q}} | \gamma \rangle}{E_q + E_{\bar{q}} - E_\gamma} \quad (2.30)$$

$$|\gamma\rangle_{q\bar{q}g}^{(3)} = \int d^3k'' d^3p'' d^3r d^3k' d^3p' d^3q' d^3k d^3p |q_{\lambda_1'}^{\alpha''}(k'') \bar{q}_{\lambda_2'}^{\beta''}(p'') g_j^b(r)\rangle \langle q_{\lambda_1'}^{\alpha''}(k'') \bar{q}_{\lambda_2'}^{\beta''}(p'') g_j^b(r)| S |q_{\lambda_1'}^{\alpha'}(k') \bar{q}_{\lambda_2'}^{\beta'}(p') g_{i'}^{a'}(q')\rangle \times \\ \frac{\langle q_{\lambda_1'}^{\alpha'}(k') \bar{q}_{\lambda_2'}^{\beta'}(p') g_{i'}^{a'}(q') | H_{qg} | q_{\lambda_1}^{\alpha}(k) \bar{q}_{\lambda_2}^{\beta}(p) \rangle}{E_{\bar{q}}(p') + E_q(k') + E_g(q') - E_{\gamma}} \times \frac{\langle q_{\lambda_1}^{\alpha}(k) \bar{q}_{\lambda_2}^{\beta}(p) | H_{\gamma \rightarrow q\bar{q}} | \gamma_{\lambda}(q) \rangle}{E_{\bar{q}}(p) + E_q(k) - E_{\gamma}(q, Q)} \quad (2.31)$$

$$|\gamma^i(w)\rangle_{q\bar{q}g}^{(3)} = \frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^{\alpha}(\zeta q^+, x) \bar{q}_{\lambda_2}^{\beta}(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\eta}{\sqrt{\xi}} \delta^{(2)}(w - c) \times \\ \left[\tilde{\Psi}_{\lambda_1 \lambda_2}^{im}[\tilde{\mathcal{G}}](\tilde{R}, X) \times [U^{ab}(z) V(x) t^b V^{\dagger}(y) - t^a]_{\alpha\beta} \right] \quad (2.32)$$

with

$$\tilde{\Psi}_{\lambda_1 \lambda_2}^{im}[\tilde{\mathcal{G}}] = \sum_{\lambda} \left[\tau_{\lambda_1 \lambda}^{nm}(\xi, \zeta) \varphi_{\lambda \lambda_2}^{ij}(\zeta + \xi) \tilde{R}^j X^n \tilde{\mathcal{G}}^{(11)} + i \xi \tilde{\omega}_f \mu_{\lambda_1 \lambda}^m \varphi_{\lambda \lambda_2}^{ij}(\zeta + \xi) \tilde{R}^j \tilde{\mathcal{G}}^{(10)} \right. \\ \left. - i m_f \tau_{\lambda_1 \lambda}^{nm}(\xi, \zeta) \mu_{\lambda \lambda_2}^i X^n \tilde{\mathcal{G}}^{(01)} + m_f \xi \tilde{\omega}_f \mu_{\lambda_1 \lambda}^m \mu_{\lambda \lambda_2}^i \tilde{\mathcal{G}}^{(00)} \right] \quad (2.33)$$

where

$$\tilde{\mathcal{G}}^{(\alpha\beta)}(\tilde{R}, X) = \frac{\tilde{a}^{\beta}}{2\eta(1-\eta)} \int_0^1 \frac{dx}{x^{1+\beta}} \left(\frac{\tilde{A}_x}{\tilde{\Delta}_x} \right)^{\frac{(\alpha+\beta)}{2}} K_{\alpha+\beta}(\sqrt{\tilde{A}_x \tilde{\Delta}_x}), \alpha, \beta \in \{0, 1\} \\ \tilde{A}_x = \tilde{\epsilon}_f^2 + \frac{x \tilde{\omega}_f^2}{\tilde{a}} = \eta(1-\eta) Q^2 + m_f^2 \left(1 + \frac{x\eta\xi}{\zeta} \right) \\ \tilde{\Delta}_x = \tilde{R}^2 + \frac{\tilde{a} X^2}{x}$$

Explicitly

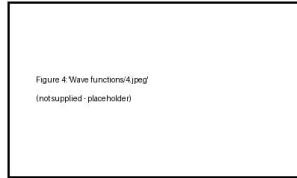
$$\tilde{\mathcal{G}}^{(11)} = \frac{\tilde{a}}{2\eta(1-\eta)} \int_0^1 \frac{dx}{x^2} \frac{\tilde{A}_x}{\tilde{\Delta}_x} K_2 \left(\sqrt{\tilde{A}_x \tilde{\Delta}_x} \right) \quad (2.34)$$

$$\tilde{\mathcal{G}}^{(10)} = \frac{1}{2\eta(1-\eta)} \int_0^1 \frac{dx}{x} \sqrt{\frac{\tilde{A}_x}{\tilde{\Delta}_x}} K_1 \left(\sqrt{\tilde{A}_x \tilde{\Delta}_x} \right) \quad (2.35)$$

$$\tilde{\mathcal{G}}^{(01)} = \frac{\tilde{a}}{2\eta(1-\eta)} \int_0^1 \frac{dx}{x^2} \sqrt{\frac{\tilde{A}_x}{\tilde{\Delta}_x}} K_1 \left(\sqrt{\tilde{A}_x \tilde{\Delta}_x} \right) \quad (2.36)$$

$$\tilde{\mathcal{G}}^{(00)} = \frac{1}{2\eta(1-\eta)} \int_0^1 \frac{dx}{x} K_0 \left(\sqrt{\tilde{A}_x \tilde{\Delta}_x} \right) \quad (2.37)$$

5. Quark emits a gluon after the shock wave



$$|\gamma\rangle_{q\bar{q}g}^{(4)} = |q_2 \bar{q}_2 g_2\rangle \langle q_2 \bar{q}_2 g_2| S |q_1 \bar{q}_1 g_1\rangle \times \frac{\langle q_1 \bar{q}_1 g_1 | H_{qg} | q\bar{q} \rangle}{E_{\bar{q}_1} + E_{q_1} - E_{q_2} - E_{\bar{q}_2} - E_{g_2}} \times \frac{\langle q\bar{q} | H_{\gamma \rightarrow q\bar{q}} | \gamma \rangle}{E_q + E_{\bar{q}} - E_{\gamma}} \quad (2.38)$$

$$|\gamma\rangle_{q\bar{q}g}^{(4)} = \int d^3k'' d^3p'' d^3r d^3k' d^3p' d^3k d^3p |q_{\lambda_1'}^{\alpha''}(k'') \bar{q}_{\lambda_2'}^{\beta''}(p'') g_j^b(r)\rangle \frac{\langle q_{\lambda_1'}^{\alpha''}(k'') \bar{q}_{\lambda_2'}^{\beta''}(p'') g_j^b(r) | H_{qg} | q_{\lambda_1'}^{\alpha'}(k') \bar{q}_{\lambda_2'}^{\beta'}(p') \rangle}{E_q(k') + E_{\bar{q}}(p') - E_q(k'') - E_{\bar{q}}(p'') - E_g(r)} \times \\ \langle q_{\lambda_1'}^{\alpha'}(k') \bar{q}_{\lambda_2'}^{\beta'}(p') | S | q_{\lambda_1}^{\alpha}(k) \bar{q}_{\lambda_2}^{\beta}(p) \rangle \times \frac{\langle q_{\lambda_1}^{\alpha}(k) \bar{q}_{\lambda_2}^{\beta}(p) | H_{\gamma \rightarrow q\bar{q}} | \gamma_{\lambda}(q) \rangle}{E_{\bar{q}}(p) + E_q(k) - E_{\gamma}(q, Q)} \quad (2.39)$$

Substituting the relations we get

$$|\gamma^i(w)\rangle_{q\bar{q}g}^{(4)} = -\frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^{\alpha}(\zeta q^+, x) \bar{q}_{\lambda_2}^{\beta}(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\eta}{\sqrt{\xi}} \delta^{(2)}(w - c) \times \\ \tilde{\Psi}_{\lambda_1 \lambda_2}^{im}[\tilde{\mathcal{A}}](\tilde{R}, X) \times [t^a V(x') V^{\dagger}(y) - t^a]_{\alpha\beta} \quad (2.40)$$

where

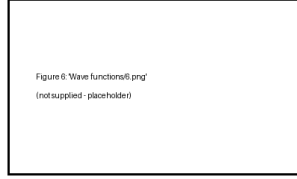
$$\tilde{\mathcal{A}}^{(11)} = \frac{\tilde{\epsilon}_f K_1(\tilde{\epsilon}_f \tilde{R})}{\eta(1-\eta) \tilde{R}} \frac{\tilde{\omega}_f K_1(\tilde{\omega}_f X)}{X} \quad (2.41)$$

$$\tilde{\mathcal{A}}^{(10)} = \frac{\tilde{\epsilon}_f K_1(\tilde{\epsilon}_f \tilde{R})}{\eta(1-\eta) \tilde{R}} K_0(\tilde{\omega}_f X) \quad (2.42)$$

$$\tilde{\mathcal{A}}^{(01)} = \frac{K_0(\tilde{\epsilon}_f \tilde{R})}{\eta(1-\eta)} \frac{\tilde{\omega}_f K_1(\tilde{\omega}_f X)}{X} \quad (2.43)$$

$$\tilde{\mathcal{A}}^{(00)} = \frac{K_0(\tilde{\epsilon}_f \tilde{R})}{\eta(1-\eta)} K_0(\tilde{\omega}_f X) \quad (2.44)$$

6. Instantaneous contribution (Quark emission of gluon when an instantaneous vertex acts before the shockwave)



$$|\gamma\rangle_{q\bar{q}g}^{(3\text{ inst.})} = |q\bar{q}g\rangle \langle q\bar{q}g| S |q\bar{q}g\rangle \frac{\langle q\bar{q}g| H_{\gamma \rightarrow q\bar{q}g} |\gamma\rangle}{E_{q\bar{q}g} - E_\gamma} \quad (2.45)$$

Using the transformations from anti quark to quark with the insertions of Wilson lines, we get

$$|\gamma\rangle_{q\bar{q}g}^{(3\text{ inst.})} = -\frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |\bar{q}_{\lambda_2}^\beta(\eta q^+, y), g_m^a(\xi q^+, z), q_{\lambda_1}^\alpha(\zeta q^+, x)\rangle \frac{\zeta \eta \sqrt{\xi}}{(1-\eta)} \delta^{(2)}(w-c) (\delta^{im} - 2i \epsilon^{im} \lambda_2) \delta_{\lambda_1 \lambda_2} \frac{E_f K_1(E_f \mathcal{D})}{\mathcal{D}} [U^{ab}(z) V(x) t^b V^\dagger(y) - t^a]_{\alpha\beta} \quad (2.46)$$

Now, we will write the quark emissions as how we wrote the anti quark section. We will use the effective vertex formalism.

$$|\gamma^i(w)\rangle_{q\bar{q}g} = \frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, x) \bar{q}_{\lambda_2}^\beta(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\eta}{\sqrt{\xi}} \delta^{(2)}(w-c) \times \left[\tilde{\mathcal{M}}_{1\lambda_1\lambda_2}^{im}(x, y, z, \eta, \xi) \times [U^{ab}(z) V(x) t^b V^\dagger(y) - t^a]_{\alpha\beta} - \tilde{\mathcal{M}}_{2\lambda_1\lambda_2}^{im}[t^a V(x') V^\dagger(y) - t^a]_{\alpha\beta} \right] \quad (2.47)$$

with the effective vertex

$$\tilde{\mathcal{M}}_{1\lambda_1\lambda_2}^{im}(x, y, z, \eta, \xi) = \tilde{\Psi}_{\lambda_1\lambda_2}^{im}[\tilde{\mathcal{G}}](\tilde{R}, X) - \delta_{\lambda_1\lambda_2} \frac{\zeta \xi}{(\xi + \zeta)} (\delta^{im} - 2i \epsilon^{im} \lambda_1) \mathcal{G}_E \quad (2.48)$$

$$\tilde{\mathcal{M}}_{2\lambda_1\lambda_2}^{im}(x, y, z, \eta, \xi) = \tilde{\Psi}_{\lambda_1\lambda_2}^{im}[\tilde{\mathcal{A}}](\tilde{R}, X) \quad (2.49)$$

3 Trijet production cross section

3.1 Direct contributions - Anti quark contribution

$$\frac{d^3 N^{\text{direct}}}{d^3 k_1 d^3 k_2 d^3 k_3} = \frac{e^2 g^2 \zeta^2 C_F N_c}{8(2\pi)^{12} \xi q^{+2}} \delta(q^+ - k_1^+ - k_2^+ - k_3^+) \int_{x,y,z,\bar{x},\bar{y},\bar{z}} e^{-i k_1 \cdot (x-\bar{x})} e^{-i k_2 \cdot (y-\bar{y})} e^{-i k_3 \cdot (z-\bar{z})} \sum_f e_f^2 \left[\mathcal{H}[\mathcal{G}; \bar{\mathcal{G}}] \mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z}) - \mathcal{H}[\mathcal{A}; \bar{\mathcal{G}}] \mathcal{W}(x, y', y', \bar{x}, \bar{y}, \bar{z}) - \mathcal{H}[\mathcal{G}; \bar{\mathcal{A}}] \mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}') + \mathcal{H}[\mathcal{A}; \bar{\mathcal{A}}] \mathcal{W}(x, y', y', \bar{x}, \bar{y}', \bar{y}') \right] \quad (3.1)$$

where

$$\mathcal{H}[F; \bar{F}] = \sum_{\lambda_1, \lambda_2} \sum_{l,r} \mathcal{M}_{\lambda_1\lambda_2}^{lr}[F](x, y, z) \mathcal{M}_{\lambda_1\lambda_2}^{lr*}[\bar{F}](\bar{x}, \bar{y}, \bar{z}) \quad (3.2)$$

- Regular $\times$ Regular

$$\begin{aligned} \mathcal{H}_{\text{reg.} \times \text{reg.}}[F; \bar{F}] = & R^i Y^m \bar{R}^j \bar{Y}^n \left\{ 8 [\eta^2 + (1 - \zeta)^2] [\zeta^2 + (1 - \zeta)^2] \delta^{ij} \delta^{mn} - 8\xi(2\zeta - 1)(2(1 - \zeta) - \xi) \epsilon^{ij} \epsilon^{mn} \right\} F^{(11)} \bar{F}^{(11)} \\ & + 8 m_f^2 [\eta^2 + (1 - \zeta)^2] (Y \cdot \bar{Y}) F^{(01)} \bar{F}^{(01)} + 8 \xi^2 \omega_f^2 [\zeta^2 + (1 - \zeta)^2] (R \cdot \bar{R}) F^{(10)} \bar{F}^{(10)} \\ & + 8 m_f^2 \xi^2 \omega_f^2 F^{(00)} \bar{F}^{(00)} \\ & - 8\zeta\eta m_f \xi \omega_f \left[(R \cdot Y) F^{(11)} \bar{F}^{(00)} + (\bar{R} \cdot \bar{Y}) F^{(00)} \bar{F}^{(11)} + (R \cdot \bar{Y}) F^{(10)} \bar{F}^{(01)} + (Y \cdot \bar{R}) F^{(01)} \bar{F}^{(10)} \right] \end{aligned} \quad (3.3)$$

- Regular $\times$ Instantaneous + h.c.

$$\begin{aligned} \mathcal{H}_{\text{reg.} \times \text{inst.}}[F; \bar{F}] = & -\frac{8\zeta\xi\eta^2}{(1 - \zeta)} \left[(R \cdot Y) F^{(11)} \bar{\mathcal{G}}_E + (\bar{R} \cdot \bar{Y}) \mathcal{G}_E \bar{F}^{(11)} \right] \\ & + \frac{8\xi^2\eta m_f \omega_f}{(1 - \zeta)} \left[F^{(00)} \bar{\mathcal{G}}_E + \mathcal{G}_E \bar{F}^{(00)} \right] \end{aligned} \quad (3.4)$$

- Instantaneous $\times$ Instantaneous

$$\mathcal{H}_{\text{inst.} \times \text{inst.}}[F; \bar{F}] = 8 \left[\frac{\xi\eta}{(1 - \zeta)} \right]^2 \mathcal{G}_E \bar{\mathcal{G}}_E \quad (3.5)$$

where:

$$\mathcal{M}_{\lambda_1 \lambda_2}^{im}[\mathcal{G}] \equiv \mathcal{M}_{1\lambda_1 \lambda_2}^{im} = \Psi_{\lambda_1 \lambda_2}^{im}[\mathcal{G}](R, Y) - \delta_{\lambda_1 \lambda_2} \frac{\xi\eta}{(1 - \zeta)} (\delta^{im} + 2i\epsilon^{im} \lambda_1) \mathcal{G}_E \quad (3.6)$$

$$\mathcal{M}_{\lambda_1 \lambda_2}^{im}[\mathcal{A}] \equiv \mathcal{M}_{2\lambda_1 \lambda_2}^{im} = \Psi_{\lambda_1 \lambda_2}^{im}[\mathcal{A}](R, Y) \quad (3.7)$$

$$\begin{aligned} \Psi_{\lambda_1 \lambda_2}^{im}[\mathcal{G}] = & \sum_{\lambda} \left[\varphi_{\lambda_1 \lambda}^{ij}(\zeta) \tau_{\lambda \lambda_2}^{mn}(\xi, \eta) R^j Y^n \mathcal{G}^{(11)} + i \xi \omega_f \varphi_{\lambda_1 \lambda}^{ij}(\zeta) \mu_{\lambda \lambda_2}^m R^j \mathcal{G}^{(10)} \right. \\ & \left. - i m_f \mu_{\lambda_1 \lambda}^i \tau_{\lambda \lambda_2}^{mn}(\xi, \eta) Y^n \mathcal{G}^{(01)} + m_f \xi \omega_f \mu_{\lambda_1 \lambda}^i \mu_{\lambda \lambda_2}^m \mathcal{G}^{(00)} \right] \end{aligned} \quad (3.8)$$

Explicitly:

$$\begin{aligned} \mathcal{G}^{(\alpha\beta)}(R, Y) = & \frac{a^\beta}{2\zeta(1 - \zeta)} \int_0^1 \frac{dx}{x^{1+\beta}} \left(\frac{A_x}{\Delta_x} \right)^{\frac{(\alpha+\beta)}{2}} K_{\alpha+\beta}(\sqrt{A_x \Delta_x}), \alpha, \beta \in \{0, 1\} \\ A_x = & \epsilon_f^2 + \frac{x \omega_f^2}{a} = \zeta(1 - \zeta) Q^2 + m_f^2 \left(1 + \frac{x\zeta\xi}{\eta} \right) \\ \Delta_x = & R^2 + \frac{a Y^2}{x} \end{aligned} \quad (3.9)$$

$$\mathcal{G}^{(10)} = \frac{1}{2\zeta(1 - \zeta)} \int_0^1 \frac{dx}{x} \sqrt{\frac{A_x}{\Delta_x}} K_1(\sqrt{A_x \Delta_x}) \quad (3.10)$$

$$\mathcal{G}^{(01)} = \frac{a}{2\zeta(1 - \zeta)} \int_0^1 \frac{dx}{x^2} \sqrt{\frac{A_x}{\Delta_x}} K_1(\sqrt{A_x \Delta_x}) \quad (3.11)$$

$$\mathcal{G}^{(00)} = \frac{1}{2\zeta(1 - \zeta)} \int_0^1 \frac{dx}{x} K_0(\sqrt{A_x \Delta_x}) \quad (3.12)$$

and

$$\begin{aligned} \Psi_{\lambda_1 \lambda_2}^{im}[\mathcal{A}] = & \sum_{\lambda} \left[\varphi_{\lambda_1 \lambda}^{ij}(\zeta) \tau_{\lambda \lambda_2}^{mn}(\xi, \eta) R^j Y^n \mathcal{A}^{(11)} + i \xi \omega_f \varphi_{\lambda_1 \lambda}^{ij}(\zeta) \mu_{\lambda \lambda_2}^m R^j \mathcal{A}^{(10)} \right. \\ & \left. - i m_f \mu_{\lambda_1 \lambda}^i \tau_{\lambda \lambda_2}^{mn}(\xi, \eta) Y^n \mathcal{A}^{(01)} + m_f \xi \omega_f \mu_{\lambda_1 \lambda}^i \mu_{\lambda \lambda_2}^m \mathcal{A}^{(00)} \right] \end{aligned} \quad (3.13)$$

Explicitly:

$$\begin{aligned}\mathcal{A}^{(11)} &= \frac{\epsilon_f K_1(\epsilon_f R)}{\zeta(1-\zeta) R} \frac{\omega_f K_1(\omega_f Y)}{Y} \\ \mathcal{A}^{(10)} &= \frac{\epsilon_f K_1(\epsilon_f R)}{\zeta(1-\zeta) R} K_0(\omega_f Y) \\ \mathcal{A}^{(01)} &= \frac{K_0(\epsilon_f R)}{\zeta(1-\zeta)} \frac{\omega_f K_1(\omega_f Y)}{Y} \\ \mathcal{A}^{(00)} &= \frac{K_0(\epsilon_f R)}{\zeta(1-\zeta)} K_0(\omega_f Y)\end{aligned}$$

Finally:

$$\mathcal{G}_E = \frac{E_f K_1(E_f \mathcal{D})}{\mathcal{D}}$$

where:

$$\mathcal{D}^2(x, y, z, \zeta, \xi) = \zeta(1-\zeta) R^2 + \frac{\xi \eta Y^2}{(1-\zeta)} = \zeta \eta (x-y)^2 + \zeta \xi (x-z)^2 + \eta \xi (y-z)^2$$

$$\mu_{\lambda\lambda'}^i = \delta_{\lambda', -\lambda} [2\lambda \delta^{i1} - i\delta^{i2}] = \chi_\lambda^\dagger \sigma^3 \sigma^i \chi_{\lambda'}$$

$$\epsilon_f^2 = \tilde{Q}^2 + m_f^2 = \zeta(1-\zeta) Q^2 + m_f^2, \quad \omega_f = \frac{\xi m_f}{(1-\zeta)}, \quad E_f^2 = Q^2 + m_f^2 \left(\frac{1}{\zeta} + \frac{1}{\eta} \right), \quad a = \frac{\xi \eta}{\zeta(1-\zeta)^2}$$

$$\varphi_{\lambda_1\lambda}^{ij}(\zeta) = \delta_{\lambda_1\lambda} [(2\zeta-1)\delta^{ij} + 2i\epsilon^{ij}\lambda_1], \quad \tau_{\lambda\lambda_2}^{mn}(\xi, \eta) = \delta_{\lambda\lambda_2} [(2(1-\zeta)-\xi)\delta^{mn} + 2i\xi\epsilon^{mn}\lambda]$$

and

$$c = \zeta x + \eta y + \xi z$$

is the centre of energy of trijet system. $\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z})$ is the Wilson line structure. This has 4 terms:

- $\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z})$

$$\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z}) = S_{q\bar{q}g\,q\bar{q}g}(x, y, z, \bar{x}, \bar{y}, \bar{z}) - S_{q\bar{q}g}(x, y, z) - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) + 1 \quad (3.14)$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(x, y, z, \bar{x}, \bar{y}, \bar{z}) = Q(x, z, \bar{z}, \bar{x}) Q(z, y, \bar{y}, \bar{z})$$

$$S_{q\bar{q}g}(x, y, z) = S(x, z) S(z, y)$$

$$S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) = S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z})$$

$$\boxed{\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z}) = Q(x, z, \bar{z}, \bar{x}) Q(z, y, \bar{y}, \bar{z}) - S(x, z) S(z, y) - S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z}) + 1}$$

- $\mathcal{W}(x, y', y', \bar{x}, \bar{y}, \bar{z})$

$$\mathcal{W}(x, y', y', \bar{x}, \bar{y}, \bar{z}) = S_{q\bar{q}g\,q\bar{q}g}(x, y', y', \bar{x}, \bar{y}, \bar{z}) - S_{q\bar{q}g}(x, y', y') - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) + 1 \quad (3.15)$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(x, y', y', \bar{x}, \bar{y}, \bar{z}) = Q(x, y', \bar{z}, \bar{x}) Q(y', y', \bar{y}, \bar{z}) = Q(x, y', \bar{z}, \bar{x}) S(\bar{y}, \bar{z})$$

$$S_{q\bar{q}g}(x, y', y') = S(x, y') S(y', y') = S(x, y')$$

$$S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) = S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z})$$

$$\boxed{\mathcal{W}(x, y', y', \bar{x}, \bar{y}, \bar{z}) = Q(x, y', \bar{z}, \bar{x}) S(\bar{y}, \bar{z}) - S(x, y') - S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z}) + 1}$$

- $\mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}')$

$$\mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}') = S_{q\bar{q}g\,q\bar{q}g}(x, y, z, \bar{x}, \bar{y}', \bar{y}') - S_{q\bar{q}g}(x, y, z) - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') + 1 \quad (3.16)$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(x, y, z, \bar{x}, \bar{y}', \bar{y}') = Q(x, z, \bar{y}', \bar{x}) Q(z, y, \bar{y}', \bar{y}') = Q(x, z, \bar{y}', \bar{x}) S(z, y)$$

$$S_{q\bar{q}g}(x, y, z) = S(x, z) S(z, y)$$

$$S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') = S(\bar{y}', \bar{x}) S(\bar{y}', \bar{y}') = S(\bar{y}', \bar{x})$$

$$\boxed{\mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}') = Q(x, z, \bar{y}', \bar{x}) S(z, y) - S(x, z) S(z, y) - S(\bar{y}', \bar{x}) + 1}$$

- $\mathcal{W}(x, y', y', \bar{x}, \bar{y}', \bar{y}')$

$$\mathcal{W}(x, y', y', \bar{x}, \bar{y}', \bar{y}') = S_{q\bar{q}g\,q\bar{q}g}(x, y', y', \bar{x}, \bar{y}', \bar{y}') - S_{q\bar{q}g}(x, y', y') - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') + 1 \quad (3.17)$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(x, y', y', \bar{x}, \bar{y}', \bar{y}') = Q(x, y', \bar{y}', \bar{x}) Q(y', y', \bar{y}', \bar{y}') = Q(x, y', \bar{y}', \bar{x})$$

$$S_{q\bar{q}g}(x, y', y') = S(x, y') S(y', y') = S(x, y')$$

$$S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') = S(\bar{y}', \bar{x}) S(\bar{y}', \bar{y}') = S(\bar{y}', \bar{x})$$

$$\boxed{\mathcal{W}(x, y', y', \bar{x}, \bar{y}', \bar{y}') = Q(x, y', \bar{y}', \bar{x}) - S(x, y') - S(\bar{y}', \bar{x}) + 1}$$

3.2 Direct contribution - Quark contributions

The following transformations are made from the anti quark sector to quark sector:

$$\zeta \rightarrow \eta, R \rightarrow \tilde{R}, Y \rightarrow X, x \leftrightarrow y$$

and

$$\epsilon_f \rightarrow \tilde{\epsilon}_f, \omega_f \rightarrow \tilde{\omega}_f, \mathcal{G} \rightarrow \tilde{\mathcal{G}}, \mathcal{A} \rightarrow \tilde{\mathcal{A}}$$

$$\frac{d^3 N^{\text{direct}}}{d^3 k_1 d^3 k_2 d^3 k_3} = \frac{e^2 g^2 \eta^2 C_F N_c}{8 (2\pi)^{12} \xi q^{+2}} \delta(q^+ - k_1^+ - k_2^+ - k_3^+) \int_{x, y, z, \bar{x}, \bar{y}, \bar{z}} e^{-i k_1 \cdot (x - \bar{x})} e^{-i k_2 \cdot (y - \bar{y})} e^{-i k_3 \cdot (z - \bar{z})} \sum_f e_f^2 \left[\tilde{\mathcal{H}}[\tilde{\mathcal{G}}; \tilde{\mathcal{G}}] \mathcal{W}(y, x, z, \bar{y}, \bar{x}, \bar{z}) - \tilde{\mathcal{H}}[\tilde{\mathcal{A}}; \tilde{\mathcal{G}}] \mathcal{W}(y, x', x', \bar{y}, \bar{x}, \bar{z}) - \tilde{\mathcal{H}}[\tilde{\mathcal{G}}; \tilde{\mathcal{A}}] \mathcal{W}(y, x, z, \bar{y}, \bar{x}', \bar{x}') + \tilde{\mathcal{H}}[\tilde{\mathcal{A}}; \tilde{\mathcal{A}}] \mathcal{W}(y, x', x', \bar{y}, \bar{x}', \bar{x}') \right] \quad (3.18)$$

where:

$$\tilde{\mathcal{H}}[F; \bar{F}] = \sum_{\lambda_1, \lambda_2} \sum_{l, r} \tilde{\mathcal{M}}_{\lambda_1 \lambda_2}^{lr} [F](x, y, z) \tilde{\mathcal{M}}_{\lambda_1 \lambda_2}^{lr*} [\bar{F}](\bar{x}, \bar{y}, \bar{z}) \quad (3.19)$$

- Regular $\times$ Regular

$$\begin{aligned} \tilde{\mathcal{H}}_{\text{reg.} \times \text{reg.}} = & \tilde{R}^i X^m \tilde{R}^j \bar{X}^n \left\{ 8 [\zeta^2 + (1 - \eta)^2] [\eta^2 + (1 - \eta)^2] \delta^{ij} \delta^{mn} - 8\xi (2\eta - 1) (2(1 - \eta) - \xi) \epsilon^{ij} \epsilon^{mn} \right\} F^{(11)} \bar{F}^{(11)} \\ & + 8m_f^2 [\zeta^2 + (1 - \eta)^2] (X \cdot \bar{X}) F^{(01)} \bar{F}^{(01)} + 8\xi^2 \tilde{\omega}_f^2 [\eta^2 + (1 - \eta)^2] (\tilde{R} \cdot \tilde{\bar{R}}) F^{(10)} \bar{F}^{(10)} + 8m_f^2 \xi^2 \tilde{\omega}_f^2 F^{(00)} \bar{F}^{(00)} \\ & + 8\zeta \eta m_f \xi \tilde{\omega}_f \left[(\tilde{R} \cdot X) F^{(11)} \bar{F}^{(00)} + (\tilde{\bar{R}} \cdot \bar{X}) F^{(00)} \bar{F}^{(11)} + (\tilde{R} \cdot \bar{X}) F^{(10)} \bar{F}^{(01)} + (X \cdot \tilde{\bar{R}}) F^{(01)} \bar{F}^{(10)} \right] \end{aligned} \quad (3.20)$$

- Regular $\times$ Instantaneous + h.c.

$$\begin{aligned} \tilde{\mathcal{H}}_{\text{reg.} \times \text{inst.}} = & \frac{8\zeta^2 \xi \eta}{(1 - \eta)} \left[(\tilde{R} \cdot X) F^{(11)} \bar{\mathcal{G}}_E + (\tilde{\bar{R}} \cdot \bar{X}) \mathcal{G}_E \bar{F}^{(11)} \right] \\ & + \frac{8\xi^2 \zeta m_f \tilde{\omega}_f}{(1 - \eta)} \left[F^{(00)} \bar{\mathcal{G}}_E + \mathcal{G}_E \bar{F}^{(00)} \right] \end{aligned} \quad (3.21)$$

- Instantaneous $\times$ Instantaneous

$$\tilde{\mathcal{H}}_{\text{inst.} \times \text{inst.}} = 8 \left[\frac{\xi \zeta}{(1 - \eta)} \right]^2 \mathcal{G}_E \bar{\mathcal{G}}_E \quad (3.22)$$

where:

$$\tilde{\mathcal{M}}_{\lambda_1 \lambda_2}^{im} [\tilde{\mathcal{G}}] \equiv \tilde{\mathcal{M}}_{1\lambda_1 \lambda_2}^{im} = \tilde{\Psi}_{\lambda_1 \lambda_2}^{im} [\tilde{\mathcal{G}}] (\tilde{R}, X) - \delta_{\lambda_1 \lambda_2} \frac{\zeta \xi}{(\xi + \zeta)} (\delta^{im} - 2i \epsilon^{im} \lambda_1) \mathcal{G}_E \quad (3.23)$$

$$\tilde{\mathcal{M}}_{\lambda_1 \lambda_2}^{im} [\tilde{\mathcal{A}}] \equiv \tilde{\mathcal{M}}_{2\lambda_1 \lambda_2}^{im} = \tilde{\Psi}_{\lambda_1 \lambda_2}^{im} [\tilde{\mathcal{A}}] (\tilde{R}, X) \quad (3.24)$$

with

$$\begin{aligned} \tilde{\Psi}_{\lambda_1 \lambda_2}^{im} [\tilde{\mathcal{G}}] = & \sum_{\lambda} \left[\tau_{\lambda_1 \lambda}^{nm} (\xi, \zeta) \varphi_{\lambda \lambda_2}^{ij} (\zeta + \xi) \tilde{R}^j X^n \tilde{\mathcal{G}}^{(11)} + i\xi \tilde{\omega}_f \mu_{\lambda_1 \lambda}^m \varphi_{\lambda \lambda_2}^{ij} (\zeta + \xi) \tilde{R}^j \tilde{\mathcal{G}}^{(10)} \right. \\ & \left. - im_f \tau_{\lambda_1 \lambda}^{nm} (\xi, \zeta) \mu_{\lambda \lambda_2}^i X^n \tilde{\mathcal{G}}^{(01)} + m_f \xi \tilde{\omega}_f \mu_{\lambda_1 \lambda}^m \mu_{\lambda \lambda_2}^i \tilde{\mathcal{G}}^{(00)} \right] \end{aligned} \quad (3.25)$$

where

$$\begin{aligned} \tilde{\mathcal{G}}^{(\alpha\beta)} (\tilde{R}, X) = & \frac{\tilde{a}^\beta}{2\eta(1 - \eta)} \int_0^1 \frac{dx}{x^{1+\beta}} \left(\frac{\tilde{A}_x}{\tilde{\Delta}_x} \right)^{\frac{(\alpha+\beta)}{2}} K_{\alpha+\beta} (\sqrt{\tilde{A}_x \tilde{\Delta}_x}), \alpha, \beta \in \{0, 1\} \\ \tilde{A}_x = & \tilde{\epsilon}_f^2 + \frac{x \tilde{\omega}_f^2}{\tilde{a}} = \eta(1 - \eta) Q^2 + m_f^2 \left(1 + \frac{x\eta\xi}{\zeta} \right) \\ \tilde{\Delta}_x = & \tilde{R}^2 + \frac{\tilde{a} X^2}{x} \end{aligned}$$

Explicitly

$$\tilde{\mathcal{G}}^{(11)} = \frac{\tilde{a}}{2\eta(1 - \eta)} \int_0^1 \frac{dx}{x^2} \frac{\tilde{A}_x}{\tilde{\Delta}_x} K_2 \left(\sqrt{\tilde{A}_x \tilde{\Delta}_x} \right) \quad (3.26)$$

$$\tilde{\mathcal{G}}^{(10)} = \frac{1}{2\eta(1-\eta)} \int_0^1 \frac{dx}{x} \sqrt{\frac{\tilde{A}_x}{\tilde{\Delta}_x}} K_1 \left(\sqrt{\tilde{A}_x \tilde{\Delta}_x} \right) \quad (3.27)$$

$$\tilde{\mathcal{G}}^{(01)} = \frac{\tilde{a}}{2\eta(1-\eta)} \int_0^1 \frac{dx}{x^2} \sqrt{\frac{\tilde{A}_x}{\tilde{\Delta}_x}} K_1 \left(\sqrt{\tilde{A}_x \tilde{\Delta}_x} \right) \quad (3.28)$$

$$\tilde{\mathcal{G}}^{(00)} = \frac{1}{2\eta(1-\eta)} \int_0^1 \frac{dx}{x} K_0 \left(\sqrt{\tilde{A}_x \tilde{\Delta}_x} \right) \quad (3.29)$$

and

$$\begin{aligned} \tilde{\Psi}_{\lambda_1 \lambda_2}^{im}[\tilde{\mathcal{A}}] = \sum_{\lambda} \left[\tau_{\lambda_1 \lambda}^{nm}(\xi, \zeta) \varphi_{\lambda \lambda_2}^{ij}(\zeta + \xi) \tilde{R}^j X^n \tilde{\mathcal{A}}^{(11)} + i\xi \tilde{\omega}_f \mu_{\lambda_1 \lambda}^m \varphi_{\lambda \lambda_2}^{ij}(\zeta + \xi) \tilde{R}^j \tilde{\mathcal{A}}^{(10)} \right. \\ \left. - im_f \tau_{\lambda_1 \lambda}^{nm}(\xi, \zeta) \mu_{\lambda \lambda_2}^i X^n \tilde{\mathcal{A}}^{(01)} + m_f \xi \tilde{\omega}_f \mu_{\lambda_1 \lambda}^m \mu_{\lambda \lambda_2}^i \tilde{\mathcal{A}}^{(00)} \right] \end{aligned} \quad (3.30)$$

Explicitly:

$$\tilde{\mathcal{A}}^{(11)} = \frac{\tilde{\epsilon}_f K_1(\tilde{\epsilon}_f \tilde{R})}{\eta(1-\eta) \tilde{R}} \frac{\tilde{\omega}_f K_1(\tilde{\omega}_f X)}{X} \quad (3.31)$$

$$\tilde{\mathcal{A}}^{(10)} = \frac{\tilde{\epsilon}_f K_1(\tilde{\epsilon}_f \tilde{R})}{\eta(1-\eta) \tilde{R}} K_0(\tilde{\omega}_f X) \quad (3.32)$$

$$\tilde{\mathcal{A}}^{(01)} = \frac{K_0(\tilde{\epsilon}_f \tilde{R})}{\eta(1-\eta)} \frac{\tilde{\omega}_f K_1(\tilde{\omega}_f X)}{X} \quad (3.33)$$

$$\tilde{\mathcal{A}}^{(00)} = \frac{K_0(\tilde{\epsilon}_f \tilde{R})}{\eta(1-\eta)} K_0(\tilde{\omega}_f X) \quad (3.34)$$

and

- $\xi \omega_f \mu_{\lambda \lambda_2}^m \rightarrow \xi \tilde{\omega}_f \mu_{\lambda_1 \lambda}^m$ with $\tilde{\omega}_f = \frac{\xi m_f}{(1-\eta)}$
- $\epsilon_f^2 \rightarrow \tilde{\epsilon}_f^2 = \eta(1-\eta) Q^2 + m_f^2$, $a \rightarrow \tilde{a} = \frac{\xi \zeta}{\eta(1-\eta)^2}$
- $A_x \rightarrow \tilde{A}_x = \eta(1-\eta) Q^2 + m_f^2 \left(1 + \frac{x\eta\xi}{\zeta} \right)$ and $\Delta_x \rightarrow \tilde{\Delta}_x = \tilde{R}^2 + \frac{\tilde{a} X^2}{x}$
- $\mathcal{G}^{(\alpha\beta)}(R, Y) \rightarrow \tilde{\mathcal{G}}^{(\alpha\beta)}(\tilde{R}, X)$ and $\mathcal{A}^{(\alpha,\beta)}(R, Y) \rightarrow \tilde{\mathcal{A}}^{(\alpha\beta)}(\tilde{R}, X)$ also, $\zeta(1-\zeta) \rightarrow \eta(1-\eta)$

$\mathcal{W}(y, x, z, \bar{y}, \bar{x}, \bar{z})$ is the Wilson line structure. This has 4 terms:

- $\mathcal{W}(y, x, z, \bar{y}, \bar{x}, \bar{z})$

$$\mathcal{W}(y, x, z, \bar{y}, \bar{x}, \bar{z}) = S_{q\bar{q}g\,q\bar{q}g}(y, x, z, \bar{y}, \bar{x}, \bar{z}) - S_{q\bar{q}g}(y, x, z) - S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}, \bar{z}) + 1 \quad (3.35)$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(y, x, z, \bar{y}, \bar{x}, \bar{z}) = Q(y, z, \bar{z}, \bar{y}) Q(z, x, \bar{x}, \bar{z})$$

$$S_{q\bar{q}g}(y, x, z) = S(y, z) S(z, x)$$

$$S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}, \bar{z}) = S(\bar{z}, \bar{y}) S(\bar{x}, \bar{z})$$

$$\boxed{\mathcal{W}(y, x, z, \bar{y}, \bar{x}, \bar{z}) = Q(y, z, \bar{z}, \bar{y}) Q(z, x, \bar{x}, \bar{z}) - S(y, z) S(z, x) - S(\bar{z}, \bar{y}) S(\bar{x}, \bar{z}) + 1}$$

- $\mathcal{W}(y, x', x', \bar{y}, \bar{x}, \bar{z})$

$$\mathcal{W}(y, x', x', \bar{y}, \bar{x}, \bar{z}) = S_{q\bar{q}g\,q\bar{q}g}(y, x', x', \bar{y}, \bar{x}, \bar{z}) - S_{q\bar{q}g}(y, x', x') - S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}, \bar{z}) + 1 \quad (3.36)$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(y, x', x', \bar{y}, \bar{x}, \bar{z}) = Q(y, x', \bar{z}, \bar{y}) Q(x', x', \bar{x}, \bar{z}) = Q(y, x', \bar{z}, \bar{y}) S(\bar{x}, \bar{z})$$

$$S_{q\bar{q}g}(y, x', x') = S(y, x') S(x', x') = S(y, x')$$

$$S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}, \bar{z}) = S(\bar{z}, \bar{y}) S(\bar{x}, \bar{z})$$

$$\boxed{\mathcal{W}(y, x', x', \bar{y}, \bar{x}, \bar{z}) = Q(y, x', \bar{z}, \bar{y}) S(\bar{x}, \bar{z}) - S(y, x') - S(\bar{z}, \bar{y}) S(\bar{x}, \bar{z}) + 1}$$

- $\mathcal{W}(y, x, z, \bar{y}, \bar{x}', \bar{x}')$

$$\mathcal{W}(y, x, z, \bar{y}, \bar{x}', \bar{x}') = S_{q\bar{q}g\,q\bar{q}g}(y, x, z, \bar{y}, \bar{x}', \bar{x}') - S_{q\bar{q}g}(y, x, z) - S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}', \bar{x}') + 1 \quad (3.37)$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(y, x, z, \bar{y}, \bar{x}', \bar{x}') = Q(y, z, \bar{x}', \bar{y}) Q(z, x, \bar{x}', \bar{x}') = Q(y, z, \bar{x}', \bar{y}) S(z, x)$$

$$S_{q\bar{q}g}(y, x, z) = S(y, z) S(z, x)$$

$$S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}', \bar{x}') = S(\bar{x}', \bar{y}) S(\bar{x}', \bar{x}') = S(\bar{x}', \bar{y})$$

$$\boxed{\mathcal{W}(y, x, z, \bar{y}, \bar{x}', \bar{x}') = Q(y, z, \bar{x}', \bar{y}) S(z, x) - S(y, z) S(z, x) - S(\bar{x}', \bar{y}) + 1}$$

- $\mathcal{W}(y, x', x', \bar{y}, \bar{x}', \bar{x}')$

$$\mathcal{W}(y, x', x', \bar{y}, \bar{x}', \bar{x}') = S_{q\bar{q}g\,q\bar{q}g}(y, x', x', \bar{y}, \bar{x}', \bar{x}') - S_{q\bar{q}g}(y, x', x') - S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}', \bar{x}') + 1 \quad (3.38)$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(y, x', x', \bar{y}, \bar{x}', \bar{x}') = Q(y, x', \bar{x}', \bar{y}) Q(x', x', \bar{x}', \bar{x}') = Q(y, x', \bar{x}', \bar{y})$$

$$S_{q\bar{q}g}(y, x', x') = S(y, x') S(x', x') = S(y, x')$$

$$S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}', \bar{x}') = S(\bar{x}', \bar{y}) S(\bar{x}', \bar{x}') = S(\bar{x}', \bar{y})$$

$$\boxed{\mathcal{W}(y, x', x', \bar{y}, \bar{x}', \bar{x}') = Q(y, x', \bar{x}', \bar{y}) - S(y, x') - S(\bar{x}', \bar{y}) + 1}$$

3.3 Interference contributions

$$\begin{aligned} \frac{d^3 N^{\text{cross}}}{d^3 k_1 d^3 k_2 d^3 k_3} &= -\frac{e^2 g^2 \eta \zeta C_F N_c}{8 (2\pi)^{12} \xi q^{+2}} \delta(q^+ - k_1^+ - k_2^+ - k_3^+) \times 2 \Re \int_{x, y, z, \bar{x}, \bar{y}, \bar{z}} e^{-i k_1 \cdot (x - \bar{x})} e^{-i k_2 \cdot (y - \bar{y})} e^{-i k_3 \cdot (z - \bar{z})} \sum_f e_f^2 \\ &\left[\mathcal{H}^x[\tilde{\mathcal{G}}; \bar{\mathcal{G}}] \mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z}) - \mathcal{H}^x[\tilde{\mathcal{A}}; \bar{\mathcal{G}}] \mathcal{W}(x', y, x', \bar{x}, \bar{y}, \bar{z}) - \mathcal{H}^x[\tilde{\mathcal{G}}; \bar{\mathcal{A}}] \mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}') \right. \\ &\left. + \mathcal{H}^x[\tilde{\mathcal{A}}; \bar{\mathcal{A}}] \mathcal{W}(x', y, x', \bar{x}, \bar{y}', \bar{y}') \right] \end{aligned} \quad (3.39)$$

where

$$\mathcal{H}^x[\tilde{F}; \bar{F}] = \sum_{\lambda_1, \lambda_2} \sum_{l, r} \tilde{\mathcal{M}}_{\lambda_1 \lambda_2}^{lr}[\tilde{F}](x, y, z) \mathcal{M}_{\lambda_1 \lambda_2}^{lr*}[\bar{F}](\bar{x}, \bar{y}, \bar{z}) \quad (3.40)$$

- Regular $\times$ Regular

$$\begin{aligned} \mathcal{H}_{\text{reg.} \times \text{reg.}}^x &= \tilde{R}^i X^m \bar{R}^j \bar{Y}^n \left\{ 8[1 - \xi - 2\zeta\eta] [\xi(1 - \xi) + 2\zeta\eta] \delta^{ij} \delta^{mn} - 8\xi[1 - 2\zeta - \xi]^2 \epsilon^{ij} \epsilon^{mn} \right\} \tilde{F}^{(11)} \bar{F}^{(11)} \\ &+ 8m_f^2 (\xi + 2\zeta\eta) (X \cdot \bar{Y}) \tilde{F}^{(01)} \bar{F}^{(01)} - 8(\xi + 2\eta\zeta) \xi^2 \tilde{\omega}_f \omega_f (\tilde{R} \cdot \bar{R}) \tilde{F}^{(10)} \bar{F}^{(10)} - 8(\zeta + \xi)^2 m_f \xi \omega_f (\tilde{R} \cdot X) \tilde{F}^{(11)} \bar{F}^{(00)} \\ &+ 8(1 - \zeta)^2 m_f \xi \tilde{\omega}_f (\bar{R} \cdot \bar{Y}) \tilde{F}^{(00)} \bar{F}^{(11)} + 8\eta^2 m_f \xi \tilde{\omega}_f (\tilde{R} \cdot \bar{Y}) \tilde{F}^{(10)} \bar{F}^{(01)} - 8\zeta^2 m_f \xi \omega_f (X \cdot \bar{R}) \tilde{F}^{(01)} \bar{F}^{(10)} \end{aligned} \quad (3.41)$$

- Regular $\times$ Instantaneous + h.c.

$$\mathcal{H}_{\text{reg.} \times \text{inst.}}^x = -\frac{8\xi \eta (\xi + \zeta)^2}{(1 - \zeta)} (\tilde{R} \cdot X) \tilde{F}^{(11)} \bar{\mathcal{G}}_E + \frac{8\xi \zeta (1 - \zeta)^2}{(\xi + \zeta)} (\bar{R} \cdot \bar{Y}) \mathcal{G}_E \bar{F}^{(11)} \quad (3.42)$$

- Instantaneous $\times$ Instantaneous

$$\mathcal{H}_{\text{inst.} \times \text{inst.}}^x = \frac{\zeta \xi}{(\zeta + \xi)} \frac{\xi \eta}{(1 - \zeta)} \mathcal{G}_E \bar{\mathcal{G}}_E \sum_{\lambda_1} \sum_{l, r} (\delta^{lr} - 2i\epsilon^{lr} \lambda_1) (\delta^{lr} + 2i\epsilon^{lr} \lambda_1)^* = 0 \quad (3.43)$$

$\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z})$ is the Wilson line structure. This has 4 terms:

- $\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z})$

$$\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z}) = S_{q\bar{q}g\,q\bar{q}g}(x, y, z, \bar{x}, \bar{y}, \bar{z}) - S_{q\bar{q}g}(x, y, z) - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) + 1 \quad (3.44)$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(x, y, z, \bar{x}, \bar{y}, \bar{z}) = Q(x, z, \bar{z}, \bar{x}) Q(z, y, \bar{y}, \bar{z})$$

$$S_{q\bar{q}g}(x, y, z) = S(x, z) S(z, y)$$

$$S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) = S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z})$$

$$\boxed{\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z}) = Q(x, z, \bar{z}, \bar{x}) Q(z, y, \bar{y}, \bar{z}) - S(x, z) S(z, y) - S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z}) + 1}$$

- $\mathcal{W}(x', y, x', \bar{x}, \bar{y}, \bar{z})$

$$\mathcal{W}(x', y, x', \bar{x}, \bar{y}, \bar{z}) = S_{q\bar{q}g\,q\bar{q}g}(x', y, x', \bar{x}, \bar{y}, \bar{z}) - S_{q\bar{q}g}(x', y, x') - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) + 1 \quad (3.45)$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(x', y, x', \bar{x}, \bar{y}, \bar{z}) = Q(x', x', \bar{z}, \bar{x}) Q(x', y, \bar{y}, \bar{z}) = S(\bar{z}, \bar{x}) Q(x', y, \bar{y}, \bar{z})$$

$$S_{q\bar{q}g}(x', y, x') = S(x', x') S(x', y) = S(x', y)$$

$$S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) = S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z})$$

$$\boxed{\mathcal{W}(x', y, x', \bar{x}, \bar{y}, \bar{z}) = S(\bar{z}, \bar{x}) Q(x', y, \bar{y}, \bar{z}) - S(x', y) - S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z}) + 1}$$

- $\mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}')$

$$\mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}') = S_{q\bar{q}g\,q\bar{q}g}(x, y, z, \bar{x}, \bar{y}', \bar{y}') - S_{q\bar{q}g}(x, y, z) - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') + 1 \quad (3.46)$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(x, y, z, \bar{x}, \bar{y}', \bar{y}') = Q(x, z, \bar{y}', \bar{x}) Q(z, y, \bar{y}', \bar{y}') = Q(x, z, \bar{y}', \bar{x}) S(z, y)$$

$$S_{q\bar{q}g}(x, y, z) = S(x, z) S(z, y)$$

$$S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') = S(\bar{y}', \bar{x}) S(\bar{y}', \bar{y}') = S(\bar{y}', \bar{x})$$

$$\boxed{\mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}') = Q(x, z, \bar{y}', \bar{x}) S(z, y) - S(x, z) S(z, y) - S(\bar{y}', \bar{x}) + 1}$$

- $\mathcal{W}(x', y, x', \bar{x}, \bar{y}', \bar{y}')$

$$\mathcal{W}(x', y, x', \bar{x}, \bar{y}', \bar{y}') = S_{q\bar{q}g\,q\bar{q}g}(x', y, x', \bar{x}, \bar{y}', \bar{y}') - S_{q\bar{q}g}(x', y, x') - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') + 1 \quad (3.47)$$

At large N_c limit, we get

$$S_{q\bar{q}g\,q\bar{q}g}(x', y, x', \bar{x}, \bar{y}', \bar{y}') = Q(x', x', \bar{y}', \bar{x}) Q(x', y, \bar{y}', \bar{y}') = S(\bar{y}', \bar{x}) S(x', y)$$

$$S_{q\bar{q}g}(x', y, x') = S(x', x') S(x', y) = S(x', y)$$

$$S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') = S(\bar{y}', \bar{x}) S(\bar{y}', \bar{y}') = S(\bar{y}', \bar{x})$$

$$\boxed{\mathcal{W}(x', y, x', \bar{x}, \bar{y}', \bar{y}') = S(\bar{y}', \bar{x}) S(x', y) - S(x', y) - S(\bar{y}', \bar{x}) + 1}$$

A Corrections made to this version

Line numbers refer to `Trijet_massive.tex` as supplied. Items 1–13 change results; items 14–30 are typos, notation or wording.

Errors that change results

1. L384, L493, L606 (all three cross sections): the factor Q^2 in the prefactor was removed. With massive radial functions the photon virtuality is already inside $\mathcal{G}, \mathcal{A}, \mathcal{G}_E$ (for $m_f = 0$, $\mathcal{G}^{(11)}\tilde{\mathcal{G}}^{(11)} = \frac{1}{Y^2\bar{Y}^2} \frac{Q^2 K_1(Q\mathcal{D}) K_1(Q\bar{\mathcal{D}})}{\mathcal{D}\bar{\mathcal{D}}}$), so keeping Q^2 in front counts it twice.
2. L384, L493, L606: $(\sum_f e_f^2)$ moved inside the coordinate integrals as $\sum_f e_f^2 [\dots]$, because every radial function depends on m_f and therefore on the flavour.
3. L177, L325, L428, L539: $\mathcal{G}^{(00)}$ and $\tilde{\mathcal{G}}^{(00)}$ had a spurious factor $\sqrt{A_x/\Delta_x}$. The general formula gives $(A_x/\Delta_x)^{(\alpha+\beta)/2} = 1$ for $\alpha = \beta = 0$, so $\mathcal{G}^{(00)} = \frac{1}{2\zeta(1-\zeta)} \int_0^1 \frac{dx}{x} K_0(\sqrt{A_x\Delta_x})$. Dimension check: $\mathcal{G}^{(00)}$ is dimensionless, while $\sqrt{A_x/\Delta_x}$ has dimension mass^2 .
4. L163, L311, L415, L525: the prefactor of the general formula is a^β (resp. $\tilde{a}^\beta$), with $a = \frac{\xi\eta}{\zeta(1-\zeta)^2}$, not α^β (the Greek letter α is the helicity-path label). Also L311, L525: argument $(R, X) \rightarrow (\tilde{R}, X)$.
5. L214: the $\tilde{p}$ transform must have $e^{-i\tilde{p}\cdot Y}$ (the phase derived in L136) to give $-iY^n$, and the result is $-\frac{iY^n}{Y^2} \omega_f K_1(\omega_f Y)$, not $-\frac{iY^n}{Y^2} \omega_f K_1(\omega_f Y)$. The latter has the wrong dimension and does not reduce to the massless $-iY^n/Y^2$ for $m_f \rightarrow 0$.
6. L217: $e^{+i\tilde{p}\cdot Y} \rightarrow e^{-i\tilde{p}\cdot Y}$, as in L136 and L156; otherwise the $\mathcal{A}^{(01)}$ term would have $+iY^n$.
7. L384: the Wilson-line arguments of the second and fourth terms were wrong: $\mathcal{W}(x, y, y', \bar{x}, \bar{y}, \bar{z}) \rightarrow \mathcal{W}(x, y', y', \bar{x}, \bar{y}, \bar{z})$ and $\mathcal{W}(x, y, y', \bar{x}, \bar{y}, \bar{y}') \rightarrow \mathcal{W}(x, y', y', \bar{x}, \bar{y}', \bar{y}')$. These are the structures actually worked out below in L459–L485.
8. L393: $F^{(02)} \rightarrow F^{(01)}$ (there is no (02) path; the term is the square of the photon-flip path).
9. L543: the after-shock-wave vertex function of the *quark* channel was written with the antiquark structure $(\varphi(\zeta)\tau(\xi, \eta), R, Y, \omega_f, \text{photon mass vertex } \mu_{\lambda_1\lambda})$. It must be $\tilde{\Psi}[\tilde{\mathcal{A}}]$, i.e. the same structure as $\tilde{\Psi}[\tilde{\mathcal{G}}]$ (L308) with $\tilde{\mathcal{G}} \rightarrow \tilde{\mathcal{A}}: \tau_{\lambda_1\lambda}^{nm}(\xi, \zeta)\varphi_{\lambda\lambda_2}^{ij}(\zeta+\xi), \tilde{R}, X, \tilde{\omega}_f$.

10. L615 (interference, Regular $\times$ Regular, $\tilde{R} \cdot \bar{R}$ term): $\xi^2 \tilde{\omega}_f^2 \omega_f \tilde{F}^{(01)} \bar{F}^{(10)} \rightarrow \xi^2 \tilde{\omega}_f \omega_f \tilde{F}^{(10)} \bar{F}^{(10)}$. Each gluon-flip vertex brings one factor $\xi \omega$, and $\tilde{R} \cdot \bar{R}$ multiplies the $(10) \times (10)$ path. As printed, the term has dimension mass^3 instead of mass^2 , and its (01) function does not match the vector $\tilde{R}$.
11. L69: the antiquark term of the photon vertex lost its σ^λ : $-\frac{\sigma \cdot p + m_f \sigma^3}{p^+} \rightarrow -\sigma^\lambda \frac{\sigma \cdot p + m_f \sigma^3}{p^+}$. L74 (which has it) is the correct form.
12. L101: in the middle expression $(\tilde{k}^2 + \omega_f^2) \rightarrow (\tilde{k}^2 + \epsilon_f^2)$. The final expression on that line was already right.
13. L248 (instantaneous, momentum space): the overall sign is $+$, not $-$; the factor $\frac{1}{(2\pi)^6}$ and the colour matrix $t_{\alpha\beta}^a$ were missing. Only then does its Fourier transform, Eq. L252, give L256 (whose $+$ sign is the correct one: it is fixed relative to the regular term by the Dirac-spinor check of the effective vertex).

Typos, notation and wording

14. L63: $E_q(k) = \frac{k^2 + m^2}{2k^+} \rightarrow \frac{k^2 + m_f^2}{2k^+}$.
15. L77: $2\lambda_1 \delta^{\lambda_1} - i\delta^{\lambda_2} \rightarrow 2\lambda_1 \delta^{\lambda,1} - i\delta^{\lambda,2}$ (Cartesian components 1, 2 of the polarisation index λ , not the helicities λ_1, λ_2).
16. L86: the reduced gluon vertex $\frac{\tau \tilde{p} + \xi \omega_f \mu}{\xi \eta q^+}$ was added. The text said that the p terms cancel but did not state the result, which is what defines the vertex of L98.
17. L96, L188, L443: $\delta^{il} \rightarrow \delta^{i1}$ (letter $l \rightarrow$ number 1).
18. L189, L444: $\epsilon_f = \tilde{Q}^2 + m_f^2 \rightarrow \epsilon_f^2 = \tilde{Q}^2 + m_f^2$.
19. L103: missing “=” at the start of the second line.
20. L206, L337: energy denominator $-E_q(k'') + E_{\bar{q}}(p'') \rightarrow -E_q(k'') - E_{\bar{q}}(p'')$.
21. L228: $\Psi_{\lambda_1 \lambda}^{im}(\mathcal{A}) \rightarrow \Psi_{\lambda_1 \lambda_2}^{im}[\mathcal{A}]$; L341: $[\tilde{A}] \rightarrow [\tilde{\mathcal{A}}]$.
22. L234–L235: “ $(b) = (a)|_{y,z \rightarrow y'}$ ” holds literally only for $m_f = 0$. The text now says what replaces it with masses ($\mathcal{G} \rightarrow \mathcal{A}$), and that the $Y \rightarrow 0$ limits still agree.
23. L248, L256, L268, L367: Fock state written as $|q \bar{q} g\rangle$ in the same order as the regular terms (notation only). Only the momentum-space L248 was changed; in the others the ordering is cosmetic and was left.
24. L258–L259: the sentence referred to the factor $\frac{Y^2}{R \cdot Y}$ of the massless effective vertex, which no longer appears; it was rephrased.
25. L287: “=” $\rightarrow$ “ $\rightarrow$ ” in the rule for φ .
26. L290: “ $D \rightarrow \tilde{Q} \tilde{R}$ ” is the massless after-shock-wave rule and uses the wrong photon scale for the quark channel; it was replaced by $\mathcal{D} \rightarrow \sqrt{\eta(1-\eta)} \tilde{R}$ (massless limit only; with masses, use $\tilde{\mathcal{A}}$).
27. L401: label “reg. $\times$ inst.” $\rightarrow$ “inst. $\times$ inst.”.
28. L393, L502: $\bar{F}^{11} \rightarrow \bar{F}^{(11)}$, $\bar{F}^{01} \rightarrow \bar{F}^{(01)}$.
29. L406, L409, L515, L518: $\mathcal{M}[\mathcal{G}] \equiv \mathcal{M}_1$, $\mathcal{M}[\mathcal{A}] \equiv \mathcal{M}_2$ (and the same with tildes) made explicit, since $\mathcal{H}$ is defined through $\mathcal{M}[F]$. L610: $\tilde{\mathcal{M}}[\tilde{F}] \rightarrow \tilde{\mathcal{M}}[\tilde{F}]$ (the bar is reserved for the conjugate-amplitude coordinates).
30. L154 “andf”, L162 “is is”, L166/L314/L434/L528/L545 “Explicilty”, L215 “some normalisation”, L237 “Instantanous”.

Checked and correct as written: the massive energy denominators (final forms), E_f^2 , the vertices φ, τ, μ , the momentum-space regular wave function L128, the Schwinger representation L160, $A_x, \Delta_x, \mathcal{G}^{(11)}, \mathcal{G}^{(10)}, \mathcal{G}^{(01)}$ and their tilde versions, $\tilde{\mathcal{A}}^{(\alpha\beta)}, \tilde{\mathcal{A}}^{(\alpha\beta)}, \mathcal{G}_E$ (L252), all coordinate-space wave functions except L543, the effective vertices $\mathcal{M}_{1,2}, \tilde{\mathcal{M}}_{1,2}$, all helicity sums $\mathcal{H}, \tilde{\mathcal{H}}, \mathcal{H}^x$ except L393 and L615, the overall signs ($-\zeta, +\eta$, interference $-\eta\zeta$), and all large- N_c Wilson-line structures. The massless generic transform L152 is correct with the $e^{+ip \cdot Z}$ shown there.